Racial and Nationality Bias in Subjective Evaluations: A Replication of Principe and Van Ours (2022)

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Abstract

We study the effect of racial bias in the newspaper ratings of professional footballers in Italian domestic football, by replicating a study by Principe and van Ours (2022). We find that, conditional on the objective performance of the footballers, black players receive a lower rating than non-black players. However, for one of the newspapers in review, biased ratings are due to favouritism towards Italian footballers. The presence of racial bias in footballers' performance evaluations is present only at the lower end of the ratings distribution, potentially showing that the bias is due to unconscious bias on the part of evaluators. We also find that pay discrimination against black players is not present within Italian football, suggesting this is due to competition between clubs able to drive discrimination out.

JEL Codes: J15, J71, L82, L83, Z2

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1. Introduction

Assessing the ability and performance of a worker, particularly where true quality cannot be fully measured objectively, is a key component of organisational management and the efficient allocation of incentives, such as promotions or monetary incentives. Although the economics literature appraises the importance of subjectively determined incentives on employee's effort and firm profitability (see Lazear (2018)), for these incentives to be effective in increasing effort, they must reward individuals performances instead of luck or some other non-performance output (Prendergast, 1999). However, it is frequently observed that individuals are rewarded due to things outside of their performance, such as oil and gas company executives being compensated through shifts in oil prices (Bertrand & Mullainathan, 2001; Davis & Hausman, 2020), or footballers being evaluated more favourably by managers and sports journalists due to goals scored by chance (Gauriot & Page, 2019).

One reason individuals are compensated for factors outside of their performance is due to evaluators often being influenced by biases when assessing or comparing performances (Coupe et al., 2018). Researchers have highlighted several forms of bias that affect the true evaluation of an individual. For example, Ginsburgh and Van Ours (2003) found that expert judges within a piano competition are biased according to the order of performance, with competitors who performed later in the order receiving higher scores. Similarly, when reviewing scientific grant applications to the National Institutes of Health, Li (2017) found reviewers were more likely to fund applications and assign higher scores to those that aligned with their own area of research. A key focus focus of the economics literature on subjective evaluations, however, is whether biases are due to favouritism towards certain groups or result from discrimination against individual characteristics.

In this paper, we review bias in the newspaper ratings of footballers in the top tier of Italian domestic football – the Serie A – focusing on the effect of race on the performance evaluations given by sports journalists. To do this, we replicate the results by Principe and Van Ours (2022), providing a thorough discussion on how it aligns with the previous literature on discrimination and biased performance evaluations, along some extensions to the original analyses. Mass media provides a unique setting for reviewing subjective evaluations. Whereas within the workplace, biased performance evaluations have a direct effect on the income and career advancement of an employee (Powell & Butterfield, 1997), the economic implications of racial bias in media is not evidently direct. Instead, the presence of sporting media in society means that bias may cause negative external effects, such as newspaper readers developing an animus against black players, underscoring the importance of identifying and addressing bias, even when there is no clear direct consequence.

The present study contributes to multiple areas within the economics literature. First, we expand on the work on bias within subjective evaluations. Much of this literature appraise the negative effects of biased evaluations within firms. For example, MacLeod (2003) suggests that employees who perceive evaluations as biased against them will reduce their effort, while Takahashi et al. (2014) suggest that biased evaluations increases the probability of workers quitting. Due to the issues that arise with such bias, researchers have attempted to understand the forms of bias in performance evaluations. Here, we focus on two specific types that have been identified by the academic literature: nationality bias and racial bias, both of which stem from favouritism. Favouritism bias refers to evaluations that are influenced by an evaluator's preference for or against a certain group, rather than reflecting true performance. Researchers within the area often find that favouritism presents a form of in-group bias, where evaluators favour individuals within who share the same race, nationality, or other similar characteristic. For example, Coupe et al. (2018) find that jury members for the FIFA the best award disproportionately favour candidates from their own country, with Dee (2005) finding that teachers are more likely to negatively evaluate pupils of a different race.

Our second contribution is to the literature on discrimination. Economists have had a keen interest in the identification and explanation of labour market disparities for decades, generally considering two main theories for discrimination – taste-based and statistical discrimination. Taste-based discrimination, as first discussed by Becker (1971 [1957]), suggests that individuals who have a preference against a group would want to avoid interaction with them. For example, an employer who has a preference against black employees will hire fewer of them. Conversely, statistical discrimination involves evaluators making decisions based on imperfect information, such that the evaluator estimates a worker's productivity based on the average productivity of their group. For example, an employer may be less likely to hire a woman due to the expectation she may be out of work due to childbirth. More recent research has extended on these theories, suggesting that statistical discrimination can be formed though incorrect beliefs about a group (Bohren et al. (2019)) or that discrimination may form due to implicit biases that the evaluator may hold

(Bertrand et al. (2005)). We appraise the importance of discussing and identifying such mechanisms behind discrimination, as the different theories of discrimination affects the optimal design of policy interventions to mitigate bias.

A major challenge faced by researchers in each area is that uncovering bias requires a large dataset in which all observable characteristics affecting an individual's outcomes or evaluations are controlled for. However, this type of data is often sparsely available. As a result, researchers often utilise data from professional sports, where individual performances can be easily observed and quantified, and data is more readily available. This form of data analysis has been used frequently within both areas in review. For example, Szymanski (2000) used data from English football to review racial wage discrimination, with Zitzewitz (2006) using data from ski jumping and figure skating to study nationality bias in the assessment of competitors.

We contribute to each area of literature by reviewing the effect that race has on the subjective performance evaluations of footballers within Italian sports media. We review Italian football for two main reasons. First, sports newspapers within Italy have a distinct feature where, after each domestic football match, they provide their evaluation of each player with a rating and a short paragraph summarising all the performances, giving us a quantitative measure of an evaluators assessments. Second, discrimination in Italian football remains a persistent issue. The Observatory on Racism in Football reported 309 incidents of racism within Italian football stadiums between 2011 and 2018 (Antonsich, 2019). This problem extends to Italian sports media as well, with newspapers Gazzetta dello Sport and Corriere dello Sport receiving criticism in recent years for racist depictions and headlines involving black footballers.[1]

Our findings suggest that newspaper ratings of footballers are racially biased against black players. However, for one newspaper, we suggest this is driven by favouritism towards Italian players rather than animus against black players. Extending our results, we find that this bias in ratings is concentrated at the lower end of the ratings distribution, suggest that the bias is due to implicit discrimination from the journalists, leading them to unconsciously rate black footballers lower. However, we are unable to distinguish the bias between implicit or taste-based discrimination. Finally, we review whether there is racial discrimination in the wages of footballers, finding no evidence of pay disparities between races at any point of the

[1] See section 3.1 for a more thorough discussion of the controversies.

wage distribution. We suggest that this is due to sufficient competition between football clubs causing potential discrimination to disappear.

The present study is structured as such. Section 2 provides a review of the existing literature on discrimination and biased subjective evaluation. Section 3 discusses the background and motivation of our study, along with a review of Principe and Van Ours (2022), followed by a discussion of our data and descriptive analyses. Section 4 provides an empirical analysis, presenting our methodology, baseline results, and robustness analysis. Section 5 presents a discussion of our findings along with areas for future research, and Section 6 gives our concluding remarks.

2. Literature Review

Identifying and understanding the presence of outcome disparities among different groups has received considerable attention in the economics literature. Much of the empirical work on group disparities focuses on documenting differences in outcomes between gender and racial groups, assessing if such differences remain present whilst holding constant observable characteristics, and evaluating the factors responsible for the disparities (Guryan & Charles, 2013). However, whilst identifying group disparities is relatively straightforward, assessing whether discrimination is the cause of the disparities is more challenging. Although discrimination likely contributes to outcome disparities between groups, distinguishing the effects from variations in preferences or underlying characteristics presents an identification problem (Lang & Spitzer, 2020).

Despite the challenges, economists have had a keen interest in the identification and explanation of discrimination on labour market outcomes. The seminal work by (Becker, 1971 [1957]) began exploring how an individual’s discriminatory preferences might interact with market participants and lead to outcome disparities within a competitive framework. He considers three models where discrimination arises from economic agents: employers, co-workers, and consumers. Central across these models is a theory of taste-based discrimination that reflects an individual's animus against a group. Specifically, an individual incurs a cost (either monetarily or through disutility) when interacting with a group they hold prejudice against. For example, in the context of employer-based taste discrimination, an employer may hire a lesser qualified white worker over a more skilled black employee due to a preference for working with white people.

Becker further analysed how market competition can reduce the level of discrimination. In his model, prejudiced employers see reduced profits by paying more for (or hiring a less qualified) worker from their preferred race. Additionally, competition from non-discriminating firms would drive discriminators out of the market, suggesting that the level of discrimination within a market is largest when competition is low. This analysis faced criticism, notably by economist Kenneth Arrow, arguing "Becker's model predicts the absence of the phenomenon it was designed to explain" (Arrow, 1972). Furthermore, economists generally avoid models of behaviour based solely on individual preferences, as Stigler and Becker (1977) would later highlight.

Following the criticisms of taste-based discrimination models, Arrow (1972) and Phelps (1972) (and later expanded by Aigner and Cain (1977) and Lundberg and Startz (1983)) developed a model based on the rational optimisation of agents with imperfect information, referred to as statistical discrimination. This form of discrimination arises when membership in a group conveys information about the individual's abilities or other characteristics, but their actual productivity cannot be perfectly observed. For example, when an employer is evaluating a potential employee where there's imperfect information, their prediction of the worker's productivity is combined from individual signals of productivity (such as those from a resume) with the average productivity of the worker's race or gender.

More recent research has expanded from these baseline theoretical frameworks to further explain how discrimination forms. Bohren et al. (2019) suggest that discrimination can arise from incorrect beliefs about a group's productivity, which they term as inaccurate statistical discrimination. This form of discrimination underlines the challenges researchers face in identifying the causal effects of discrimination. The presence of inaccurate beliefs leads to an identification problem, where discrimination can be misclassified as taste-based rather than statistical. This creates an issue regarding the design of effective policy interventions that mitigate bias, as the differing forms of discrimination affect the optimal policy response. For example, addressing inaccurate statistical discrimination may involve policies to improve the information about the true distribution of abilities. Similarly, welfare analyses and the response of competitive markets to discrimination will vary depending on its form (Bohren et al., 2019; Fang & Moro, 2011). Although, considering inaccurate statistical discrimination creates challenges with identification, Bohren et al., (2019) suggest that collecting information on the subjective beliefs of evaluators, along with observing their decisions, allows for the correct identification of inaccurate beliefs over preferences. Bertrand et al. (2005) propose a similar model of discrimination, where biases can arise from implicit biases held by individuals, which may lead to unintentional discrimination from the evaluator. This is referred to as implicit discrimination. Like Bohren et al., (2019), they outline several tests to identify implicit discrimination, stating how experiments can be implemented to overcome the identification problem. However, the reliance on experimental data to for identification may lead to issues with external validity or high research costs, suggesting why there has been little focus in these mechanisms when discussion the cause of discrimination. Despite the extensions in models of discrimination, the main two remain; taste-based models where discrimination arises due to prejudice towards a group and statistical discrimination, where group membership signals an individual's ability when an evaluator cannot perfectly observe their ability.

As noted earlier, much of the economics literature on group disparities and discrimination has traditionally been focused on documenting differences in outcomes between groups, conditional on their abilities, rather than distinguishing between different models of discrimination. A recent survey of the literature on discrimination by Bohren et al. (2019) finds that just under half of studies attempt to test for the source of discrimination (whether taste-based or statistical), while around 62% discussed if the discrimination they found was preference based or statistical. Instead, empirical research on discrimination has largely focused on two main areas. One such focus is examining the effects of policies on disparities in labour market outcomes among different groups. While the role of discrimination within these disparities provides significant context, the literature is primarily concerned with how policies impact these group disparities (Guryan & Charles, 2013). Another strain of the literature attempts to measure the effect of discrimination on gaps between different groups, typically through wage gaps, by utilising regression-based methods to test for the existence of discrimination.

Empirical work that aims to measure the extent of discrimination in group outcomes typically start with a simple regression model including a binary variable that indicates an individual's race. The parameter estimates then indicate how an individual's labour market outcomes would differ if they were, for example, black instead of white, holding everything else equal, or simply, the effect of discrimination on the groups labour market outcomes. A prominent extension to this simple regression-based method attempts to decompose the gap in outcomes into an explained part based on differences in observed attributes, such as years of experience, and an unexplained part, which is usually attributed to discrimination. Multiple methods exist that achieve this decomposition. Most commonly, researchers use a linear approach as suggested by Blinder (1973) and Oaxaca (1973). However, some studies have proposed non-parametric approaches (Mora, 2008) or quantile-based decomposition (Firpo et al., 2018). In either case, the analysis seeks to measure the causal effect of being, for example, black on economic outcomes.

The use of regression-based methods to infer the causal effect of group membership has faced criticism due to both empirical and conceptual concerns. Researchers have highlighted that such studies tend to shift from causal inferences to a correlation of an individual's race or gender with outcome disparities. Gaebler et al. (2022) suggest that there are three significant challenges for estimating causal effects. Firstly, defining the causal estimands of interest becomes an issue when the treatment variable is an immutable characteristic. Conceptually, this issue arises as causal inference involves assessing the effect of altering an aspect of the world in some way (Holland, 2003). Consequently, researchers suggest it is inappropriate to conceptualise an immutable characteristic as a treatment in an observational study, since such characteristic is not changeable through intervention. Additionally, because these characteristics are determined at conception, any observable characteristics are effectively post-treatment.[2] The authors caution that adjusting for covariates determined downstream from group assignment, such as race, can lead to post-treatment bias of estimates. Huber and Solovyeva (2020) highlight how such empirical analysis (including decompositions) have previously failed to address this endogeneity, as observed characteristics are determined later in life, or can be influenced by the existence of discrimination. In other terms, for a racial wage gap, it may be minority workers invest less in education due to the anticipation of later discrimination (Altonji & Blank, 1999). As such, the unexplained part of the decomposition doesn't correspond to the true causal mechanisms.

Finally, estimates face significant challenges with omitted-variable bias if researchers don't adjust for all relevant covariates. Despite the conceptual and empirical difficulties of analysing discrimination in a causal framework, Guryan and Charles (2013) highlight how Becker's definition of discrimination – the difference in pay between two workers of equal productivity – provides a basis for empirical testing. They suggest that this distinction means

[2] For a thorough discussion on the conceptual challenges of using immutable characteristics as treatment variables, see Greiner and Rubin (2011).

that it is hypothetically feasible to test for discrimination empirically, even if it doesn't necessarily fit into a causal framework.

The criticisms surrounding regression-based methods for measuring discrimination has led to alternative methods which are now frequently used in the literature; namely, audit and correspondence studies. In-person audit studies, often referred to simply as audit studies, are a type of field experiment where researchers randomise certain characteristics about an individual and send a trained assistant into the field to observe the effects of those characteristics on specific outcomes. For instance, Neumark et al. (1996) conduct an audit study investigating gender discrimination by sending individuals to apply for restaurant jobs, finding that female applicants had a lower probability of receiving a job offer. Similarly, Wienk et al. (1979) review racial discrimination in the housing market by sending individuals to a real estate agency to simulate house searching. Conversely, correspondence studies create hypothetical individuals though mediums such as resumes or job applications to test for discrimination in responses. A widely regarded example of this is by Bertrand and Mullainathan (2004), who sent resumes that were randomly assigned a black or white sounding name, finding that those with white sounding names received 50% more callbacks than the black sounding counterpart. This type of experiment improves on some of the criticisms of regression-based analysis and allows for strong claims of causality, as the researchers can match individuals on observable characteristics other than their race, gender, or other characteristic, providing a way to hold relevant attributes constant. Whereas regression-based methods measure outcomes as an object of discrimination, audit studies examine the behaviour of the individuals possibly perpetrating discrimination. It is then possible to perceive discrimination as a difference in behaviour by employers, rather than in the outcomes of workers. This behaviour shows the importance of considering how employers then perceive race, allowing for the possibility to assign race randomly through assigning the perception of race randomly (Guryan & Charles, 2013).

Although the use of audit and correspondence studies has provided a useful advancement to the empirical toolbox when measuring the effects of discrimination, important criticisms have been highlighted that researchers are still working on. Gaddis (2018) considers three main methodological issues with audit designs. Firstly, there's a debate as to whether researchers should use paired (where two related datasets are compared) or unpaired (where two independent datasets are compared) designs when running the audit study. Second, the signal that researchers use in audit designs to convey an individual's race (or other immutable characteristic) is difficult to distinguish from a signal about other characteristics which are accounted for in determining an outcome. This presents a significant challenge due to an observer possibly believing that people with, for example, distinctively black names are distinguished by other productivity relevant attributes, such as the individuals social class instead of race (Gaddis, 2017). Due to this, it may be the result that differential behaviour is due to beliefs about the other traits instead of the race (Charles & Guryan, 2011). Gaddis (2018) highlights that findings from audit studies are strongly linked to the names used by researchers for the experiment, suggesting that results are biased if the signals used in audits aren't examined. Finally, a prominent critique by Heckman (1998) and Heckman and Siegelman (1993) suggests that researchers of an audit study assume unobservable characteristics have the same distribution across groups, but this cannot be validated. Heckman suggest that multiple components could enter the decision process which is unknown to researchers and so aren't included in the audit design. This presents an issue if the two groups have different variances on the unobserved characteristics, as the findings will under/overstate the effect of discrimination.

Much of the literature on discrimination now utilise the methods from audit studies when disentangling the causal effects of discrimination on outcomes, despite some of the highlighted flaws. Although the use of regression-based methods perhaps has the greatest limitations when inferring the causal effect of discrimination, they are still of use when understanding outcome differentials, particularly in the context of equilibrium labour market discrimination, where the use of experimental methods is limited. For the scope of this study, it is sufficient to use a regression-based method to understand the racial disparities in performance evaluations, however, the criticisms highlighted in this section show how we need to be careful to interpret outcome disparities as a causal effect of an individual's race, even if it could be suggested it is due to discrimination through Becker's definition.

The present study expands on the literature on discrimination by reviewing discrimination in the context of biased performance evaluations. Subjective evaluations are present in various contexts, such as when a police officer determines the punishment to hand out at a traffic stop, a consumer or expert providing a review of a product, or a supervisor assessing an employee's performance. While performance evaluation in the workplace is one example, this form of assessment extends to other areas. For example, in competitions, a panel of judges scores competitors, with referees in sporting events similarly making calls on fouls given their subjective evaluation. In many firms, subjective performance measures are used alongside objective metrics to inform decisions on budget allocations, investments, and employee compensation (Prendergast, 1999). Specifically, regarding employee compensation, economists appraise the role performance evaluations have in workplace incentives, particularly where objective measures of performance are rarely used. For example, promotions and bonuses tend to be allocated from the subjective evaluation of supervisors over incentives that are objectively determined, such as piece rates. This form of compensation receives a fair amount of scholarly attention, as researchers have highlighted how they are more commonly implemented. For instance, MacLeod (2003) notes that, in the 90s, around 20 to 30 percent of workers within the United States received a form of subjectively determined rewards, compared to the 1 to 5 percent that received compensation in the form of a piece rate or commission.

However, a significant issue with subjective evaluations is that they are inherently discretionary and not enforceable by a third party, which can lead to untruthful or biased performance assessments (Bouwens et al., 2022). This bias can negatively impact both employee welfare and firm performance. Prendergast (1999) argues that for incentive contracts to effectively increase worker effort, they must reward actual performance rather than luck or other non-performance factors. Similarly, MacLeod (2003) suggests that employees may reduce their effort if they perceive that evaluations of them are biased. Regarding employee welfare, Takahashi et al. (2014) finds that biased evaluations increase the likelihood of workers who face such bias quitting, with Powell and Butterfield (1997) highlighting that biased evaluations have a direct and negative impact on employee income and career advancement. Biased performance evaluations can also present issues of discrimination influencing an evaluators assessment. This immediately leads to issues due to fairness and legality for discriminated employees, however, there are also direct consequences for the firm, as Johnson and Neumark (1997) and Neumark and McLennan (1995) find that minority workers that report experiences of discrimination are more likely to leave their employers. Despite the negative consequences, some scholars have suggested that behavioural theory predicts that some forms of bias can have a positive effect on employee motivation and/or performance, if the bias increases the perceived fairness of incentives (Bol, 2011).

Researchers within the area of subjective evaluations have highlighted several forms of bias in performance evaluations. Prendergast (1999) highlights two forms of bias that are effectually similar – leniency bias and centrality bias. Leniency bias refers to the tendency of evaluators to be reluctant to given poor ratings to workers, presenting a form of rating inflation. Centrality bias, on the other hand, is the case where supervisors compress ratings around some value rather than properly distinguishing between good and bad performances. The presence of this bias is backed by some empirical evidence. For example, Bol (2011) and Moers (2005) find evidence of both leniency and centrality bias in their analysis of performance evaluations but highlight different implications. Moers (2005) suggests that the number of performance measures for supervisors, that are both objective and subjective, leads to performance ratings to be more lenient and less differentiated amongst employees. Consequently, Bol (2011) finds that the negative consequences of biased performance evaluations were due to centrality bias instead of leniency bias, suggesting that it negatively affects performance improvement. Conversely, she finds that there is no evidence of a significant and negative relationship between leniency bias and performance, instead finding evidence of behavioural theory where the increased perceived fairness of the evaluation due to the leniency increases employee's performance. Prior work has also found that subjective evaluations are prone to being biased by outside unrelated performance measures, referred to as a spill-over bias. Bol and Smith (2011) discuss this form of rating distortion, suggesting that supervisors distort information about employee performance to be consistent with more objective measures from a separate task.

Finally, empirical researchers have found evidence of order bias within performance evaluations. This refers to the tendency of evaluators to rate individuals who appear first or earlier in a sequence of performances worse than those who appear later. This isn't usually interpreted as a conscious decision that evaluators make, instead it can be argued that this is the result of judges adjusting their ratings as they can better compare the performances of individuals, or that tiredness through the competition means judges are looser with their scoring. Ginsburgh and van Ours (2003) make a similar argument to the first point as they find that the order of appearances within a piano competition affects the final ratings of the competitors. They suggest this is attributed to the jury needing to become familiar to the new play composed for the competition. This form of order bias has further been documented across different competitions, such as in music competitions (Bruine De Bruin, 2005; Flôres & Ginsburgh, 1996), gymnastics (Damisch et al., 2006; Plessner, 1999), and figure skating (Bruine De Bruin, 2005). This is likely the least consequential form of bias within a workplace, instead highlighting a possible issue to be addressed within competition designs, however, it can likely be generalised to job hiring, where an evaluator goes through resumes and interviews in turn.

Although the aforementioned forms of bias offer useful insight into distortions in subjective evaluations, they are less relevant to the present study than the other form of bias highlighted by the literature - favouritism bias. Prendergast and Topel (1996) discuss the implications of favouritism in performance evaluations, highlighting that the key implication of subjective assessments is how they enable supervisors to exercise bias or favouritism towards employees based on personal preferences over actual performance. This observation has inspired a wide range of empirical literature aimed at detecting favouritism bias in various contexts. For instance, when reviewing scientific grant applications to the National Institutes of Health, Li (2017) finds that reviewers of these applications are biased towards those that are closest to their research field.

An important finding across the literature on favouritism bias is that evaluators tend to favour individuals who share similar characteristics, such as nationality or race, this is referred to as in-group bias. Nationality bias is a frequently observed type of in-group bias that is seen in various settings, where evaluators assess individuals of the same nationality more favourably. For example, when reviewing the Eurovision song contest, Ginsburgh and Noury (2008) and Spierdijk and Vellekoop (2009) find that ratings of competitors from the panel of judges take account of factors such as the linguistic, cultural, and geographical proximity of the competitor, rather than the true quality of the performance. More generally, researchers have found that judges show favouritism to individuals with the same nationality in combat fighting (Myers et al., 2006), dressage (Sandberg, 2018), Olympic diving (Emerson et al., 2009), figure skating (Zitzewitz, 2006), football (Pope & Pope, 2015)**,** gymnastics (Heiniger & Mercier, 2021), ski jumping (Krumer et al., 2022; Zitzewitz, 2006), and voting in the FIFA the best award (Coupe et al., 2018). Favouritism bias has also been shown to manifest through gender, where evaluators tend to assess individuals of the opposite gender lower. Dee (2005) highlights this, finding that teachers are more likely to evaluate pupils of the opposite gender as more disruptive and inattentive. Similarly, Goldin and Rouse (2000) find that judges at the U.S. symphony orchestra auditions were around 50% more likely to advance a female candidate from the preliminary round when the audition were blind, indicating substantial underlying bias against female auditioners.

Finally, racial bias is another form of in-group bias frequently observed in the subjective evaluations' literature, and it serves as the primary focus of the present study. Racial bias in subjective evaluations has received considerable scholarly attention at least since Dejung and Kaplan (1962), who find black army recruits would receive higher peer ratings from members of their own race in the U.S. army. A later meta-analysis by Kraiger and Ford (1985) found that early literature on the race effects on subjective evaluations shows that white evaluators would favour white individuals in their ratings, with black evaluators showing the same bias in the evaluations of their black counterparts. A more recent study within the economics literature is provided by Elvira and Town (2001) reviews this bias using data from a U.S. company, but find a slightly different effect with black supervisors rating white employees significantly lower than black employees, but do find white managers evaluate white employees significantly higher than black employees. The authors also review pay differentials by race but find that there are no differences between black and white workers when controlling for job title and subjective evaluations, showing the direct effect racially biased evaluations has on the compensation of employees. Further empirical evidence documents a similar type of racial bias uncovered by Elvira and Town (2001), with evaluators providing harsher evaluations of the opposite race in arrest rates Donohue and Levitt (2001)**,** baseball (Parsons et al., 2011), basketball (Price & Wolfers, 2010), football (Magistro & Wack, 2023), grocery stores (Giuliano et al., 2011), legal claims (Shayo & Zussman, 2011), and pupil evaluations (Dee, 2005).

A large challenge faced by researchers reviewing discrimination and subjective evaluations is that empirical analysis needs a large dataset where all characteristics that may affect group outcomes can be controlled for and, specifically in the context of performance evaluations, comparable decisions where opinions can be quantified and bias observed is needed (Zitzewitz, 2006). However, it is often too costly to collect the necessary data, or it is sparsely available in the workplace setting. Due to this, of the economics literature utilises data from professional sports, where data is more widely available, and the performance of individuals can be easily quantified. For instance, Szymanski (2000) uses data from English professional football to study pay discrimination between black and non-black footballers, finding that black players were discriminated against in their wages. Furthermore, much of the previously referenced literature reviews biased subjective evaluations using sporting contests, where competitors are often scored by a set of judges or referees make evaluations about fouls committed by athletes, providing a useful setting for analysis. The paper the present study replicates by Principe and Van Ours (2022) follows this type of analysis, using data from Italian football to control for the performances of individuals being evaluated, they review the ratings of footballers in sports newspaper to investigate whether the evaluations are influenced by a racial bias, along with reviewing pay discrimination for black players. This study follows the focus of Elvira and Town (2001), with an expansion to a new context.

3. Background and Data

3.1 Motivation and Background

In May of 2024, FIFA President Gianni Infantino announced the governing bodies plan to tackle racial abuse in professional football, announcing the '*Global Stand Against Racism*' plan.[3] Although previous efforts have been in place to combat the presence of racism within professional sport, this recent campaign shows the persistence of racism, with Italian football highlighting the extent of racial abuse within the sport. For instance, the Observatory on Racism in Football found between 2011 and 2018, there were 309 recorded incidents of racism within Italian football stadiums (Antonsich, 2019). This issue has been further proliferated by Italian sports media. In 2012, after Italy defeated England in the UEFA European Championship, the sports newspaper Gazzetta dello Sport, the highest sold Italian sports newspaper, published a cartoon that depicted the black Italian footballer Mario Balotelli as King Kong on top of Big Ben.[4] More recently, in 2019 another sports newspaper, Corriere dello Sport, was criticised for a front page with the headline 'Black Friday' alongside photos of black footballers Romelu Lukaku and Chris Smalling.[5] An interesting feature of Italian sports newspapers is the feature of report card style ratings of footballers. These ratings were popularised by Gazzetta dello Sport, when the newspaper began publishing them in 1972. Nowadays, all sport newspapers publish these types of ratings after each domestic league football match. The ratings are assigned by the subjective opinion of sports journalists, who assign each footballer a number from 0 to 10, along with a few lines of text, to assess the performance of every player. These performance ratings act as the main

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[3] See https://inside.fifa.com/social-impact/campaigns/no-discrimination/news/fifa-congress-unites-in-support-of-strengthened-anti-racism-measures

[4] See https://www.telegraph.co.uk/sport/football/competitions/euro-2012/9358481/Euro-2012-Gazzetta-dello-Sport-apologise-for-Mario-Balotelli-King-Kong-cartoon.html

[5] See https://www.bbc.co.uk/sport/football/50668979

source of player performance information for readers where all players are evaluated, however, they are also used for player ratings in Italian fantasy football, a virtual game where players are given a budget to create a team of Serie A players, and score points from the performance of their chosen players during the match week.

Italian sports media provides a unique context for reviewing subjective evaluations. Whereas biased performance evaluations have a direct impact on the earnings and future positions of employees within a firm, we would not expect footballers who receive biased evaluations to be impacted due to it. The authors of the original study, Francesco Principe and Jan van Ours note this, highlighting that, though there are little direct economic implications for footballers, the presence of sports media in society can cause adverse spillover effects with consumers potentially developing animus against black players due to the bias. Further, they suggest that reviewing potential discrimination in media shows importance in its own right.

As noted, the study by Principe and Van Ours (2022) reviews racial bias in the newspaper ratings of Italian sports newspapers towards black footballers. They collect the ratings from the three main sports newspaper in Italy, La Gazzetta dello Sport (from now Gazzetta), Il Corriere dello Sport (from now Corriere), and TuttoSport. Further they review the wages of footballers to investigate potential pay discrimination. The authors find that, conditional on the objective performance of the footballers, that the newspaper ratings rate black players lower. However, they find that this bias is only present at the lower end of the ratings distribution. The results suggest the presence of an in-group racial bias against the black footballers, as every journalist for each newspaper in the period covered are white and Italian. Finally, they find no evidence of racial discrimination regarding pay.

The present study attempts to replicate the study by Principe and Van Ours (2022) in order to validate the findings of the original study and provide an in depth discussion about the results and how it expands the understanding of bias within subjective evaluations. Alongside this, we also expand some of the sensitivity analyses of the original paper to confirm the robustness of the original results.

3.2 Data

To replicate the original study, we use the dataset provided by the authors Francesco Principe and Jan van Ours.[6] This data records information about professional footballers in the topflight of Italian domestic football over a 9-season period, from the 2009/10 season to the 2017/18 season. Goalkeepers are excluded from the analysis following the usual approach within the (see Carrieri et al., 2020), as the performance of goalkeepers are measured differently to outfield players. After goalkeepers are excluded, we have a longitudinal dataset with 1835 player-season observations. Given the promotion/relegation and player transfer systems within professional football, the turnover of players in the league makes the panel data unbalanced. This may present concerns about selective attrition if such turnover is considered endogenous, however, the turnover of footballers involves players that are less talented being relegated or transferred to smaller clubs, along with more talented players who are transferred to larger clubs, so we do not consider player turnover to be endogenous and are not concerned about selective attrition.

The data is collected from several sources. First, the ratings of footballers are collected from the end of season overall average rating of each player that is provided by the three newspapers, Gazzetta, Corriere, and TuttoSport. Second, the yearly wages of each player, which excludes performance bonuses, are collected from an annual report on player salaries by Gazzetta. This report is published at the beginning of the league season, meaning players that join the league halfway through the season are not accounted for. For this reason, the number of observations for the analysis of player wages is 1627. Finally, measures of player performance and other observable characteristics are collected from the football statistics website www.whoscored.com. The full list of variables, along with summary statistics are provided by Table A in the appendix.

A frequent issue in the literature on racial discrimination is the establishment of an individual’s race. Charles and Guryan (2011) highlight the taxonomical challenge of race coding, arguing that built into the notion of race is the problem that the analyst’s presumption about the agent’s race may differ from what other believe about that agent. To overcome this issue, the method used by Principe and van Ours (2022) follows the approach of Grogger

[6] The replication materials for Principe and Van Ours (2022) can be found here: https://datarepository.eur.nl/articles/dataset/Racial_Bias_in_Newspaper_Ratings_of_Professional_Football_Players/17122475

(2011) by recruiting four individuals to classify a player as black or non-black after looking at a photo of that footballer.[7] The codification of the dummy variable indicating race then takes the value of one if at least one person labels the player as black. This overcomes the issue of differing perceptions of race the evaluation of race by an evaluator is based on the appearance of the individual (Palacios-Huerta, 2014). However, this method does present some concerns on the validity of player classification through over sensitivity of labelling players as black. For instance, as the identification of a player as black depends on at least one individual classifying a player as such, the coding of the variable accounts for people who misclassify a player as black. To ensure that our results are robust considering this potential oversensitivity, we later run a sensitivity analysis that uses different classifications of a player's race.

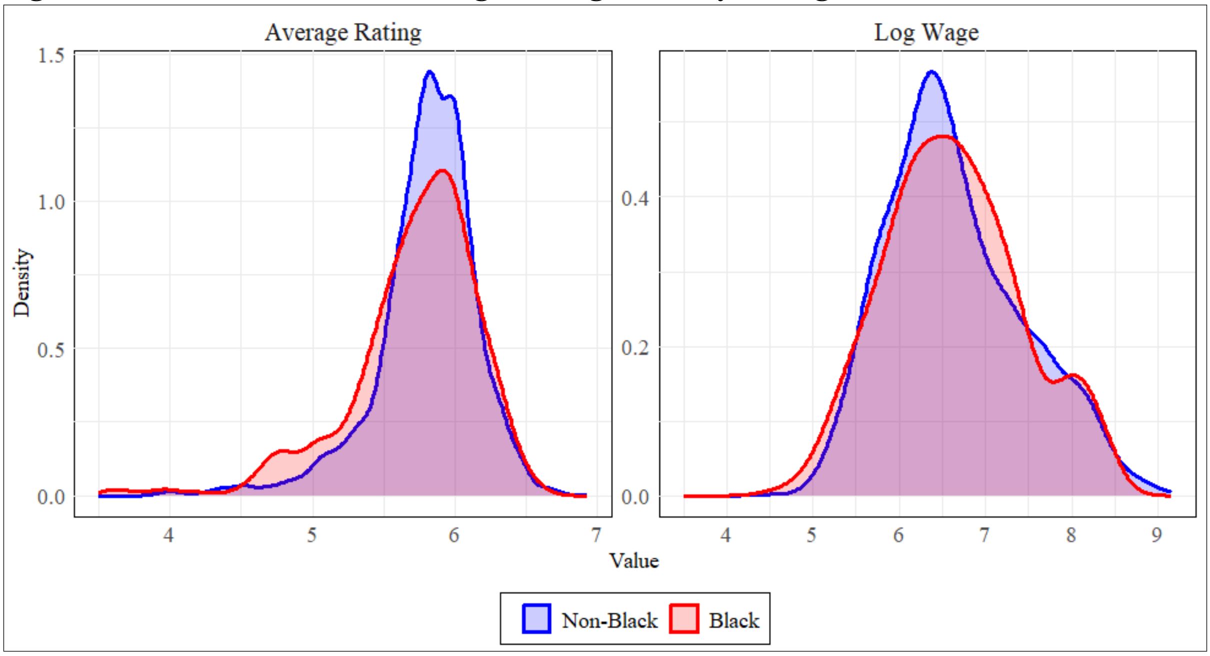
Finally, when reviewing the provided dataset, an issue that was highlighted was that values were mis inputted as 0. This is believed to be an error with data handling instead of actual ratings, as within the same observation, other ratings would be non-zero. This presents potential concerns of statistical bias in results as it is accounted for in the analysis and lowers the variable denoting average rating for observations where there is a rating error. Similarly, if the zero ratings were not in fact due to an error, they still present issues as they are significant outliers. Due to this, we run a sensitivity analysis to investigate the impact of the error on results, however, to preview, we find the effect as minimal.

3.3 Descriptive Analysis

Figure 1 presents the kernel densities of the newspaper ratings and log wages for black and non-black players. The densities within the left-hand figure show the distribution of player wages between the two groups, where we see a fairly large difference between the densities for black and non-black players. Notably, there's a relatively small deviation in ratings for non-black players around a mean rating of 5.78. In comparison, there is a slightly larger spread in the ratings of black footballers, with a lower mean rating of 5.69, where black players receive a higher frequency of lower ratings, particularly between 4.5 and 5. Conversely, the right-hand figure shows the distributions of wages between races. We see that they follow a closer distribution to each other when compared to the distribution of ratings, with little difference between the densities.

[7] Photos of the footballers were gathered from www.transfermarkt.com.

Figure 1 - Kernal Densities - Average Rating and Player Wage - Black and Non-Black



Looking closer at the distributions in player ratings, figure 2 shows the kernel densities for black and non-black ratings by newspaper, where we see a clear difference in the distribution of ratings between newspapers. The distribution of ratings by group is most similar for Gazzetta, where there is little spread in the ratings just below 6, although black players receive lower ratings slightly more frequently. The relatively small spread in ratings may suggest that the evaluations from Gazzetta is influenced by a centrality bias, suggesting that such ratings do not necessarily reflect the true quality of the performances. We discuss this point further in section 5. The distribution in ratings from Corriere and TuttoSport are similar, with a higher spread in ratings compared to Gazzetta. However, it is more noticeable that black players receive a higher frequency of lower ratings than their non-black counterparts. Table 1 provides more detail to these observations, presenting the mean ratings by race and newspaper. The table shows the mean difference in ratings is lower and statistically significant for black players across newspapers, although the size is smaller for Gazzetta.

Figure 2 - Kernal Densities by Newspaper - Black and Non-Black

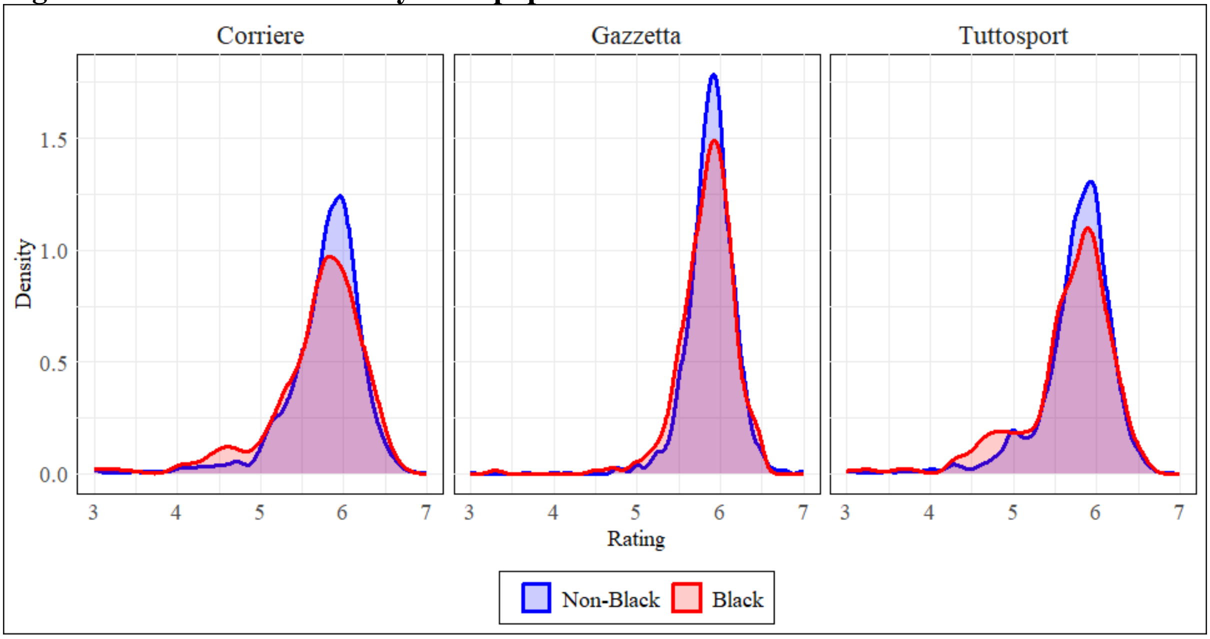


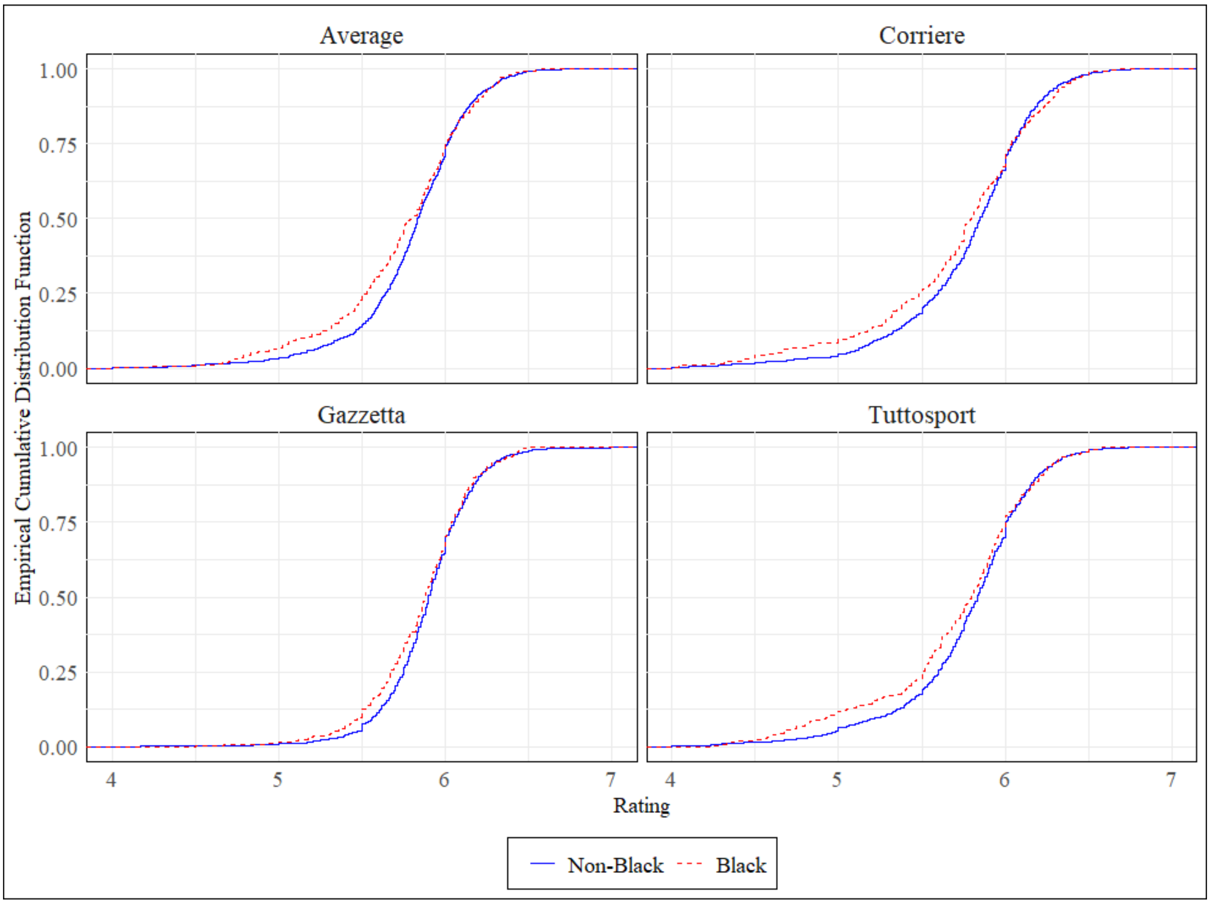
Table 1 - Mean Newspaper Rating and Player Wage: Black and Non-Black

		Black	Non-Black	Difference
Ratings	All	5.687	5.780	-0.093 ***
	Corriere	5.630	5.741	-0.111 **
	Gazzetta	5.828	5.884	-0.056 **
	TuttoSport	5.603	5.714	-0.111 ***
Log Wages		6.634	6.644	-0.010

Note: * - ** - ***: Significant at 10 - 5 - 1%

To more clearly review the differences between racial groups across the ratings distribution, figure 3 presents the empirical cumulative distribution functions for each newspaper. Across newspapers, we see the difference in ratings between groups is larger at the lower end of the distributions, again with the ratings difference smallest for Gazzetta. The figures and table from this analysis highlights that, unconditional on the quality of the player, black footballers tend to receive lower ratings than their non-black counterparts. However, this appears to be mainly present at the lower end of the ratings distribution.

Figure 3 - Empirical Cumulative Distribution Function by Newspaper - Black and Non-Black



4. Empirical Analysis

4.1 Methodology

Our analysis takes a pooled cross-section approach to empirically test if the performance evaluations of footballers in Italy depends on the players race, conditional on their on-field performance. Specifically, we estimate the following equation:

$$Y_{i,t} = \alpha + \beta_1 B_i + \beta_2 P'_{i,t} + \beta_3 X'_{i,t} + \delta_t + \tau_t + \varepsilon_{i,t} \quad (1)$$

Where $Y_{i,t}$ represents the end-of-season average rating and log wage of player i in season t, B_i is a dummy variable that indicates if at least one out of four reviewers labelled a photograph of the footballer as black, $P'_{i,t}$ and $X'_{i,t}$ are vectors that control for the player performance and personal characteristics of each player respectively, δ_t and τ_t represent season and team fixed effects, and $\varepsilon_{i,t}$ is the error term. Although the dataset provides us with panel data for the seasons in observation, our variable of interest, the player's race, does not vary over time. Due to this, we cannot employ an individual fixed-effects approach to control for unobserved differences across players. Instead, the pooled cross-sectional nature is accounted for by clustering the standard errors at the player-level.

4.1.1 Ordinary Least Squares and Unconditional Quantile Regression

We estimate equation (1) using several methods. Firstly, the baseline estimation to analyse the conditional mean difference between racial groups follows an ordinary least squares (OLS) approach. Extending the analysis, we investigate how the extent of racial bias changes across the distribution of newspaper ratings by estimating equation (1) using the unconditional quantile regression (UQR) set out by Firpo et al. (2009). Quantile regressions have been widely used within the economics literature due to the use in understanding the differential impact of different covariates along the distribution of an outcome variable. For example, when reviewing the effects of schooling on income, the method accounts for the likelihood that the impact of an additional year of schooling for individuals in the 30th percentile of the income distribution may significantly differ from an individual in the 90th percentile. The more commonly used quantile regression framework has been the conditional quantile regression (CQR) proposed by Koenker and Bassett (1978). The aim of CQR is to estimate the impact of a covariate within a quantile of a dependent variable, conditional on the specific values of other predictors. However, effect estimates from CQR may be difficult to interpret when the effects vary for different conditional quantiles (Borah & Basu, 2013). To overcome these limitations, the UQR approach has been more recently adopted. This approach estimates the effect of changing a predictor on the whole distribution of the outcome variable, without conditioning on the values of other predictors. As noted, we follow the method proposed by Firpo et al. (2009), who base the UQR method upon the concepts of an influence function (IF) and recentred influence function (RIF). The IF is a tool that is used to assess the effect of adding or removing an observation on the value of a statistic without requiring to recalculate it. The RIF is then obtained by adding the statistic to its IF (Borah & Basu, 2013). In the case where the statistic of interest is a quantile of the outcome distribution,

$$IF(y; q_\tau) = \frac{\tau - 1\{Y \leq \ q_\tau\}}{f_Y(q_\tau)} \tag{2}$$

where q_τ is the τ^{th} quantile of the player ratings, $f_Y(q_\tau)$ is the estimated density of ratings in the quantile, and $\tau - 1\{Y \leq q_\tau\}$ is an indicator variable that denotes whether an observations value is less than q_τ or not. As such, the RIF is:

$$RIF(y; q_\tau) = q_\tau + \frac{\tau - 1\{Y \leq q_\tau\}}{f_Y(q_\tau)} \tag{3}$$

An RIF regression can be considered an UQR when the conditional expectation of the RIF is modelled as a function of the independent variables (Firpo et al., 2009). The implementation of the UQR is then straightforward. In the context of Principe and Van Ours (2022) and the following analysis, the first step is to estimate the RIF for a given observed quantile q_τ from the data using equations (2) and (3). q is estimated by computing a sample quantile and then estimating the density of the unconditional distribution of the outcome (ratings and player wages in our case) using kernel density methods. Second, we run an OLS regression using the computed RIF as the dependent variable on the set of observed covariates defined in equation (1). The unconditional quantile of ratings q_τ can then be obtained as:

$$q_\tau = E_x[E(\widehat{RIF}(Y; q_\tau)|X)] \tag{4}$$

where $\widehat{RIF}(Y; q_\tau)$ is the estimate of the RIF from equation (3) conditional on X. We can then apply the law of iterated expectations due to the linear approximation, such that:

$$q_\tau = E_x[E(\widehat{RIF}(Y; q_\tau)|X)] = E[X](\hat{\delta}_\tau) \tag{5}$$

where $\hat{\delta}_\tau$ is our unconditional quantile regression coefficient. Because of the linearization, the estimation of the marginal effect of a change in the distribution of X on the unconditional quantile of our outcome variable is measured by $\hat{\delta}_\tau$.

4.1.2 Oaxaca-Blinder Decomposition and RIF Oaxaca-Blinder Decomposition

We follow the baseline estimation by implementing the Oaxaca-Blinder (OB) decomposition outlined by Blinder (1973) and Oaxaca (1973). The OB decomposition is a statistical method often used in applied economics to analyse the differences in outcomes between two groups by decomposing the difference into a part that arises from differences in observed characteristics between the groups and a part due to unknown associated factors, which is usually attributed to discrimination. The OB decomposition operates within a regression framework to determines the factors contributing to the mean differences in an outcome between two groups. In the context of Principe and Van Ours (2022), the OB decomposition decomposes the difference in the newspaper ratings and wages between black (B) and non-black (NB) players as such:

$$Y_{NB} - Y_B = \underbrace{(\bar{X}_{NB} - \bar{X}_B)\beta_B}_{Explained} + \underbrace{\bar{X}_B(\beta_{NB} - \beta_B)}_{Unexplained} \qquad (6)$$

where Y represents the average newspaper rating of wage of the player. The explained part of equation (6) is the mean difference in the levels of explanatory variables, which we consider the difference in observed characteristics or abilities of the players, and the unexplained part shows the difference in the size of regression coefficients, which is broadly attributed to discrimination.

Finally, to analyse the distributional gap in the ratings and wages between races, we utilise the flexibility of the UQR framework and apply it to the OB decomposition, as outlined by **Firpo et al. (2018)**. The difference in parameter estimates for the RIF estimation in each quantile can be decomposed as such:

$$\widehat{RIF}\left(Y_{NB}, q_{NB,\tau}\right) - \widehat{RIF}\left(Y_B, q_{B,\tau}\right) = (\bar{X}_{NB} - \bar{X}_B)\delta_B + \bar{X}_B(\delta_{NB} - \delta_B) \qquad (7)$$

4.2 Estimation Results

Table 3 presents the parameter estimates for β_1 for each of our chosen dependent variables derived from the OLS and UQR estimation Equation (1). This estimate is considered the effect of being black on a players newspaper rating and wage.

Table 2 – Effect of 'Black' on Newspaper Ratings and Player Wages - OLS and UQR

		OLS	q10	q30	q50	q70	q90
Ratings	All	-0.085 ***	-0.230 ***	-0.073 ***	-0.019	0.010	0.034
		(0.030)	(0.068)	(0.027)	(0.021)	(0.023)	(0.035)
Ratings	Gazzetta	-0.059 **	-0.105 ***	-0.058 ***	-0.023	-0.004	-0.019
		(0.030)	(0.037)	(0.021)	(0.019)	(0.021)	(0.026)
	Corriere	-0.096 **	-0.239 ***	-0.059	-0.032	0.003	0.079 **
		(0.040)	(0.075)	(0.036)	(0.026)	(0.026)	(0.040)
	TuttoSport	-0.102 ***	-0.333 ***	-0.103 ***	-0.024	-0.012	0.035
		(0.038)	(0.100)	(0.034)	(0.027)	(0.025)	(0.032)
Log Wages		-0.005	-0.049	-0.025	0.075	-0.029	0.178
		(0.050)	(0.070)	(0.063)	(0.064)	(0.103)	(0.160)

Note: Standard errors in parentheses; Standard errors are clustered at the individual player level; * - ** - ***: Significant at 10 - 5 - 1%; Full parameter estimates for all explanatory variables are presented within the appendix.

The first column of Table 3 presents the OLS estimates. We find that, after controlling for a player's characteristics and sporting performance, the average rating is 0.085 points lower for black footballers compared to their non-black counterparts, where the estimate is significant at the 1% level. We see this result across estimates for all individual newspapers, with Corriere and TuttoSport having similar coefficient sizes of -0.096 and -0.102, each significant at the 5% and 1% level respectively. The ratings bias by Gazzetta is lower than the other two newspapers, with a coefficient of -0.059. Finally, there is no evidence that there is a significant difference between player wages.

Given the results from our analysis, we validate the findings by Principe and Van Ours (2022). Figure 4 presents the OLS parameter estimates for each of the dependent variables for the original study and the replication results from Table 3. We see across specifications that the point estimates from the replication are the same as the original study. We also see slight differences in the estimation of the standard errors, however, this is likely due to differences the estimation processes between statistical software.[8]

[8] We also see the standard error for Corriere in the original paper is much larger than what we estimate. This is due to an error in the results table in the publication of the original paper, which is corrected within their appendix.

Figure 4 - Parameter Estimates - Original and Replication

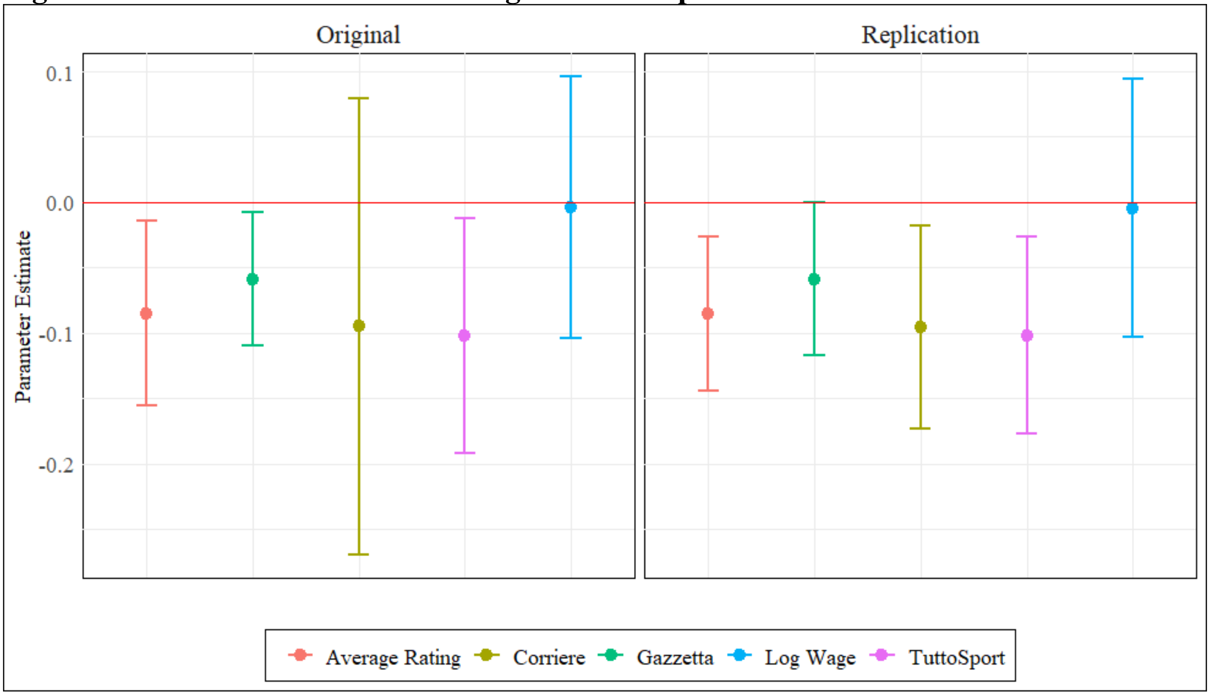


Table B in the appendix presents the parameter estimates for all explanatory variables within the analysis. The results for the variables that relate to an individuals performance show an expected effect on the newspaper ratings and player wages. Firstly, a players age has a positive but non-linear effect on player wages but not on their newspaper ratings. Actions that have a direct effect on the outcome of a football match, notably goals and assists, have a positive and significant effect across newspaper ratings and their wages. Similarly, some offensive skills such as the number of passes and crosses per game, along with the number of offensive dribbles, have a positive effect on a players rating and their wage. However, other offensive actions which may have been expected to increase a player rating, such as the number of shots on target, are not significant across newspapers specifications. Perculiarly, a higher pass success rate has a negative effect on a players rating, though this is likely that riskier passes which has a lower probability of completion are more influential in leading to assists than safer passes. Defensive skills have a similar effect on ratings as offensive skills do, with more tackles, interceptions, clearances, and blocks increasing a players rating. Finally, forward players receive lower ratings than both midfielders and defenders but receive higher wages.

The other columns within Table 3 present the parameter estimates for different quantiles from the unconditional quantile regression. We see across ratings that racial bias is present in the lower 10^{th} and 30^{th} quantiles but not statistically significant in the other parts of the ratings distribution. Further, the size of the coefficient is noticably larger in the 10^{th} quantile at -0.230 than at the 30^{th} quantile at -0.073. This effect is present across newspapers, with the exception of Corriere, where there is no racial bias in the 30^{th} quantile but a positive effect for black players in the 90^{th} quantile, with a coefficient estimate of 0.079. Again, we find no evidence that black players receive a lower wage than non-black players, with no significant estimate across the wage distribution.

To conclude the baseline analysis, Table 4 presents the Oaxaca-Blinder decompositions for our set of dependent variables. Like Table 3, the first column shows the decomposition for the OLS parameter estimates. The absolute difference between black and non-black ratings are the same as the difference in means that are presented within Table 1. Again, we see the difference in ratings between black and non-black players is statistically different from zero across newspapers. The parameter estimates for the explained gap in ratings are not significant for any newspapers, however, the unexplained component estimates are across all newspapers, with the coefficient sizes being larger than the actual differences.

Table 3 - Effect of 'Black' on Ratings and Player Wages - Oaxaca-Blinder Decomposition

		Mean	q10	q30	q50	q70	q90
All	Difference	-0.093 **	-0.280 ***	-0.118 ***	-0.059 *	-0.011	0.026
		(0.038)	(0.093)	(0.040)	(0.040)	(0.034)	(0.038)
	Explained	-0.023	-0.009	-0.029	-0.014	-0.034 *	-0.012
		(0.025)	(0.054)	(0.023)	(0.020)	(0.019)	(0.025)
	Unexplained	-0.139 ***	-0.271 ***	-0.089 ***	-0.045	0.022	0.038
		(0.041)	(0.086)	(0.033)	(0.030)	(0.029)	(0.033)
Corriere	Difference	-0.111 **	-0.332 **	-0.102 **	-0.061	-0.007	0.069
		(0.050)	(0.155)	(0.051)	(0.039)	(0.039)	(0.045)
	Explained	-0.035	-0.014	-0.037	-0.027	-0.031	-0.029
		(0.032)	(0.059)	(0.030)	(0.024)	(0.022)	(0.024)
	Unexplained	-0.162 ***	-0.318 **	-0.065	-0.035	0.024	0.098 **
		(0.047)	(0.142)	(0.042)	(0.033)	(0.034)	(0.039)
Gazzetta	Difference	-0.056 *	-0.088 ***	-0.059 **	-0.025	-0.005	-0.030
		(0.029)	(0.033)	(0.027)	(0.026)	(0.027)	(0.031)
	Explained	-0.007	-0.004	0.000	-0.006	-0.008	-0.001
		(0.018)	(0.022)	(0.018)	(0.016)	(0.017)	(0.025)
	Unexplained	-0.103 **	-0.084 **	-0.059 **	-0.019	0.003	-0.029
		(0.042)	(0.034)	(0.023)	(0.022)	(0.025)	(0.029)
Tutto Sport	Difference	-0.111 **	-0.336 ***	-0.110 **	-0.042	-0.027	0.027
		(0.049)	(0.105)	(0.043)	(0.037)	(0.035)	(0.042)
	Explained	-0.026	-0.008	-0.035	-0.037 *	-0.032	-0.021
		(0.031)	(0.084)	(0.028)	(0.022)	(0.021)	(0.026)
	Unexplained	-0.152 ***	-0.327 ***	-0.075 *	-0.005	0.005	0.048
		(0.050)	(0.096)	(0.039)	(0.032)	(0.033)	(0.036)
Log Wage	Difference	-0.010	-0.136	0.001	0.088	0.054	0.039
		(0.081)	(0.092)	(0.093)	(0.101)	(0.121)	(0.173)
	Explained	-0.007	-0.009	0.018	0.011	0.109	-0.198
		(0.075)	(0.053)	(0.068)	(0.084)	(0.121)	(0.129)
	Unexplained	0.005	-0.127	-0.016	0.077	-0.055	0.237
		(0.046)	(0.087)	(0.069)	(0.072)	(0.095)	(0.155)

Note: Standard errors in parentheses; Standard errors are clustered at the individual player level; * - ** - ***: Significant at 10 - 5 - 1%

The following columns show the parameter estimates for the Oaxaca-Blinder decomposition based on the recentred influence function. The estimates are as expected given the results from the unconditional quantile regression from Table 3. First, we see across specifications that the absolute difference is significant and negative at the lower end of the ratings distribution but is not from the 50th quartile (apart from the average rating which is significant in the 50th quartile at the 10% level). Again, we see that the explained component across each quartile is insignificant, with the exceptions of the 70th quartile for the rating average and 50th quartile for TuttoSport which are significant and negative at the 10% level. As expected, the unexplained components are significant at the lower 10th and 30th quantiles, though there is a significant positive effect for Corriere in the 90th quantile. However, unlike the mean difference, the size of the coefficients is smaller than the absolute difference. Finally, we again find that there is no effect of a players race on wages in the Oaxaca-Blinder decomposition.

4.3 Robustness Analysis

To support the robustness of our baseline findings, we run a set of sensitivity analyses. We begin by replicating the robustness analysis set out in the original paper, with an extension on our analysis of various racial subsets. Next, I review the sensitivity of the codification of a players race by employing more stringent classifications to label a player as black. Finally, I re-run our baseline analysis accounting for ratings values of zero to review the extent they bias the original results.

We begin by establishing if the bias within ratings is in fact due to the subjective judgment of journalists by re-running our original specification with an ‘objective’ rating generated by a model based on player performance. The player performance ratings are collected from www.whoscored.com. Like our newspaper ratings, we use the end-of-season average rating for each player as our dependent variable. We present this result in Table 6; alongside the baseline OLS estimate for the ratings average for comparison. We find no significant difference in the player performance rating between races, highlighting that the bias uncovered in our baseline analyses is driven by the subjective evaluation of the journalists. This result aligns with the findings in the original paper.

Table 4 - Effect of 'Black' on Newspaper Ratings and Player Performance Rating

	Newspaper Rating	Player Performance
Black	-0.085 ***	-0.018
	(0.030)	(0.013)
N	1835	1771

Note: Standard errors in parentheses; Standard errors are clustered at the individual player level; * - ** - ***: Significant at 10 - 5 - 1%

We continue the analysis by reviewing the extent distinguishing between the number of people who labelled a player as black affects our results. For this, we replace the original dummy variable, indicating if at least one person labelled the player as black, with a set of dummy variables indicating the number of people who labelled the player as black. The parameter estimates for this analysis is presented in Table 7. An important observation is that all the coefficient estimates are negative, supporting the finding of racial bias in ratings. However, we see that the effect is incremental based off the perception of the players race, with coefficient sizes increasing with the number of people labelling the player as black. However, this is only significant at the 1% level when 3 people label the player as black. The results provided suggest that the coding of a player's race may be oversensitive, leading to an overestimation of the effect of race on performance evaluations, as a single identification is not statistically significant.

Table 5 - Degree of 'Blackness'

	Number of People Labelling the Player as Black			
	1	2	3	4
All Ratings	-0.082	-0.147	-0.199 ***	-0.057 *
	(0.051)	(0.099)	(0.061)	(0.034)
Corriere	-0.080	-0.142	-0.238 ***	-0.074
	(0.058)	(0.113)	(0.066)	(0.058)
Gazzetta	-0.085	-0.086	-0.157 ***	-0.013
	(0.057)	(0.089)	(0.051)	(0.025)
TuttoSport	-0.080	-0.214 *	-0.202 ***	-0.083
	(0.056)	(0.118)	(0.076)	(0.053)
Number of Players	28	7	8	45

Note: Standard errors in parentheses; Standard errors are clustered at the individual player level; * - ** - ***: Significant at 10 - 5 - 1%

To ensure that the codifying of the original race variable doesn't bias results due to potential oversensitivity, we re-run the analysis with a new set of dummy variables indicating different criteria for the coding of a player's race (such as three of more people labelling the player black). The estimates presented in Table 8 suggest that the original classification of a player's race may be oversensitive and that a more stringent classification of a player's race may be preferred. However, across classifications the coefficient estimates are significant and negative, supporting our finding that there is a racial bias within the ratings of Italian footballers. Interestingly, when we change the classification of the black dummy variable, there is no significant presence of racial bias from Gazzetta, which may suggest that a biased estimate may lead us to falsely conclude racial bias in the newspaper's ratings.

Table 6 - Classification of a Players Race

	Dummy Variable Codification			
	1 or More	2 or More	3 or More	4
All Ratings	-0.085 ***	-0.080 **	-0.067 **	-0.040
	(0.030)	(0.032)	(0.031)	(0.034)
Corriere	-0.096 **	-0.098 **	-0.088 *	-0.056
	(0.040)	(0.050)	(0.053)	(0.058)
Gazzetta	-0.059 **	-0.034	-0.025	0.002
	(0.030)	(0.025)	(0.024)	(0.025)
TuttoSport	-0.102 ***	-0.108 **	-0.088 *	-0.064
	(0.038)	(0.046)	(0.047)	(0.053)

Note: Standard errors in parentheses; Standard errors are clustered at the individual player level; * - ** - ***: Significant at 10 - 5 - 1%

The final sensitivity analysis conducted by the original paper reviews whether racial bias in ratings is driven by favouritism towards Italian players. The authors note, 3% (7 players) are classified as black and Italian, whereas, in the sample of non-Italians, 37% (81 players) are classified as black, suggesting potential influence of Italian favouritism in ratings. To review this, they run the original estimation on a subsample of non-black players, with a new variable indicating if the player is Italian. To further this analysis, they review the effect of being black using a sample of non-Italian players. The second and third rows from Table 9 show the results from these subsamples. We find a positive, but weakly significant effect of being Italian in the sample of non-black players, across newspapers, however, there is a significant and positive effect of Italian favouritism for Gazzetta, suggesting that the finding of racial bias from this newspaper is solely due to Italian favouritism instead of racial bias. Regardless, racial bias is still present within the sample of non-Italian players across the other newspapers, suggesting that, even when we account for Italian favouritism, racial bias still influences player ratings.

Extending beyond this, I also analyse the effect of a players race by using a sample of only Italian players. By doing this, we can review whether the racial bias is driven by an ‘out-group’ bias for non-Italian players, as all the sports journalists for each newspaper are white Italians. We see in the bottom row of Table 9 that there is no significant racial bias within the sample of Italian players, although the coefficient estimates are negative across newspapers, supporting the idea that the ratings bias is driven by an out-group bias against non-Italian players. However, given the small number of black Italian players, the result may not accurately depict the effect of bias, so we are cautious in concluding that black Italians do not receive biased evaluations.

Table 7 - Subsamples based on Nationality and Ethnicity

Sample	Parameter	All	Corriere	Gazzetta	TuttoSport	N
Baseline	Black	-0.085 ***	-0.096 **	-0.059 **	-0.102 ***	1835
		(0.030)	(0.040)	(0.030)	(0.038)	
Non-black	Italian	0.038 *	0.010	0.045 **	0.059 *	1488
		(0.022)	(0.026)	(0.022)	(0.032)	
Non-Italian	Black	-0.074 **	-0.091 **	-0.047	-0.085 *	866
		(0.035)	(0.040)	(0.034)	(0.046)	
Italian	Black	-0.091	-0.231	-0.015	-0.025	969
		(0.066)	(0.240)	(0.033)	(0.096)	

Note: Standard Errors in parentheses; Clustered at the individual player level; * - ** - ***: Significant at 10 - 5 - 1%

As a final robustness check, I re-run the baseline analysis with an adjusted dataset that removes observations where a rating takes the value of zero. This reduces the number of observations for the average rating dependant variable the most, as average ratings that take the mean value of less than three newspapers are dropped. Table 10 compares the baseline results to the adjusted results, where we see that the estimates are approximately the same in terms of size and significance, however, the coefficient estimates are slightly smaller. The results validate the original findings even with the error in our dataset.

Table 8 - Estimates After Removing Values of Zero

	All	Corriere	Gazzetta	TuttoSport
Baseline Results	-0.085 ***	-0.096 **	-0.059 **	-0.102 ***
	(0.030)	(0.040)	(0.030)	(0.038)
Adjusted Results	-0.064 ***	-0.073 **	-0.039 **	-0.077 ***
	(0.024)	(0.033)	(0.020)	(0.033)
N	1828	1831	1833	1829

Note: Standard Errors in parentheses; Clustered at the individual player level; * - ** - ***: Significant at 10 - 5 - 1%; The baseline results are based on 1835 observations.

Overall, the sensitivity analyses confirm and expand on the baseline findings, along with validating the results in the original paper. We confirm that the newspapers we review show a degree of racial bias within their evaluations of footballers, however, for Gazzetta, it is likely that the ratings bias is misattributed to racial bias, as the sensitivity analysis suggests it is due to Italian favouritism instead. We further verify that the bias in newspaper ratings is due to the subjective evaluation of the journalists by analysing our model with a rating based on solely player performance, where the estimate is insignificant.

Finally, we analyse the extent the classification of a player's race affects our findings. We find, that regardless of how many people label the player as black, there is a ratings penalty for black players, however, it is only highly significant when three individuals label the player as black. However, re-estimating the model with different classifications for the race of the player suggests that the original classification of a player's race is oversensitive, highlighting that a different classification may be more suitable.

5. Discussion and Future Research

5.1 Results and Implications

Conceptually, subjective performance evaluations reflect the evaluators perceived quality of an individual's performance, less any errors in their assessments. When reviewing performance evaluations, we consider the systematic error in evaluators judgments as a bias. The presence of racial bias within evaluations need not be due to explicit discrimination on the part of the evaluators, instead an evaluation is influenced by racial bias if they don't reflect just the quality of the performance, but also the race of the individual. The results from our analysis show that the subjective evaluation of footballers from sports journalists in Italy are influenced by a degree of racial bias against black footballers. More specifically, conditional on the performance of the footballers, black players receive a lower mean rating than their non-black counterpart. This difference between races is present across all newspapers in review, also the size of the bias is heterogeneous. Specifically, our baseline analysis finds the estimated bias from Gazzetta is about half the size of the other newspapers. Extending from this, we find that the difference in ratings between racial groups is not present across the entire ratings distribution, instead, it is only present in the lower 10th and 30th quartiles. Finally, we review the influence of race on the wages of footballers, however, we find no significant differences between black and non-black players.

The racial bias we uncover provides further evidence that subjective evaluations are influenced by a form of in-group bias, as the journalists employed by the newspapers in our review are all white Italians. Although it would enhance future research to understand if the presence of in-group biases remains for evaluators of different races within this context, in the case of Italian sporting media this would not be easily examined, as the ratings provided by each newspaper are anonymously reported. Regardless, our results align with much of the literature that document the presence of this type of bias in terms of race. The findings most closely align with the results by Elvira and Town (2001) and Dee (2005), who document that the evaluation of employees and students respectively are influenced by in-group racial bias. We also support the findings of papers that identify this type of bias within similar sporting contexts, such as Magistro and Wack (2023) who find racially biased evaluations in football, Parsons et al. (2011) in baseball, and Price and Wolfers (2010) in basketball. However, the focus of these studies is slightly different, reviewing bias in foul calls, where we review performance ratings.

Following the definition of discrimination given by Gary Becker, where a difference in outcomes between workers of equal productivity amounts to discrimination, the racial bias we observe is considered discriminatory. We are unable to properly identify the mechanisms behind this discrimination with the dataset we use, although several hypotheses may be drawn. The nature of the evaluations published by the newspapers means that the performance of each player being evaluated is perfectly observed by the journalists. Necessarily, this rules out the possibility of statistical discrimination by the evaluators as this type of discrimination arises when there are information asymmetries. We then may consider the racial bias observed in evaluations as taste-based discrimination, where journalists evaluate black players lower due to an animus against them. However, it is not particularly clear whether the bias is due to the journalists' preferences, if it was, we may expect that the presence of racial bias would exist across the board; instead, we find that it is only present at the lower end of the ratings distribution. A possible explanation of this, that is presented by the authors of the original study, is that this is due to unconscious bias towards black players due to stereotyping. As racism against black individuals is still a prevalent issue within Italian football,[9] it is likely that sporting journalists implicitly hold unconscious racial animus around black players. This hypothesis is supported by the controversies that arose with the racist depiction of a black footballer by Gazzetta in 2012 and the racist headline by Corriere in 2019, as both newspapers defended their actions, further suggesting that they were not racist.[10] This suggests that the journalists are unaware of racial biases they may hold and may

[9] See section 3.1 for a discussion on this point.

[10] In a statement about the matter, Gazzetta dello Sport apologised for the depiction, but further stated "… those who accuse Gazzetta of racism are going overboard" (The Guardian, 2012). Conversely, Corriere dello Sport

portray, likely leading to their evaluations of footballers performances to be implicitly biased. Although the hypothesis that the racial bias within the newspapers ratings is due to implicit discrimination is strong, we are unable to definitively state that this is mechanism behind the bias due to the limitations of our dataset.

It is important to consider that another limitation of our analysis, and a general limitation of the regression-based framework to study outcome differentials, is that that omitted variables may overstate the effect of discrimination. Within our analysis, we are unable to control for characteristics that journalists may put an unequal weighting on which black and non-black players may have comparatively different levels of, such as leadership ability or control of the game. Although this is usually a concern with research focusing on discrimination, the use of sports data allows us to more precisely control for characteristics that influence an individual's outcome, as performance is more easily quantifiable, and observable compared to performance metrics within the workplace.

Although the focus of the present and replicated studies concerns bias in subjective evaluations, we also review potential pay discrimination against black footballers. Across our baseline analyses we don't find evidence that there are significant salary disparities in terms of race. This finding differs from the results presented by Szymanski (2000), who suggests that English football clubs discriminate in terms of pay and hiring against black players. This may indicate that salary discrimination within football has been decreasing over the previous decades, as Szymanski (2000), reviews wage discrimination between the years of 1978-93. This suggestion is supported by Kahn (2009), who notes that previous research reviewing wage discrimination within American basketball found evidence of pay disparities in the 1980s, but that the evidence was less strong when reviewing the 1990s and 2000s. Similarly, when reviewing racial discrimination within English football, Chu et al. (2014) found no evidence of salary discrimination. Again, we are only able to hypothesise why discrimination isn't present with our data, but the authors of the replicated study suggest two possible explanations. First, Italian football clubs are not discriminatory in terms of wages, suggesting that talent acquisition at football clubs don't have racial preferences for players. The second, and most likely explanation, is that there is sufficient competition in football such that discrimination amongst players is removed. This explanation follows the model of

defended their headline stating "An innocent title, […], is transformed into poison by those who have poison inside them" (Zazzaroni, 2019).

discrimination outlined by Becker (1971 [1957]), suggesting that in highly competitive markets discriminating firms will be driven out as they have a cost disadvantage compared to non-discriminating firms. In this context, the market for footballers is likely highly competitive, as football clubs compete against each other to sign the highest performing players, inefficient player acquisition due to racial bias may cause poor sporting performance, leading to decreased levels of profits or sporting success.

Uncovering racial bias within the context of the newspaper ratings of footballers provides some interesting implications when reviewing subjective evaluations and generalising the result to the workplace. First and most generally, we provide further evidence that performance evaluations are prone to discriminatory bias. From a managerial perspective, this highlights that the use of mainly subjective metrics when evaluating employees' performances and designing incentive contracts is inefficient and may lead to issues such as incentive contracts not being effective (Prendergast, 1999), employees reducing their effort (MacLeod, 2003), or employees having an increased probability of quitting (Takahashi et al., 2014). Conversely, within professional sports, it highlights potential issues of fairness where scoring or outcomes are determined subjectively, which may mean reforms that change the scoring systems of the sporting contests must be implemented to ensure fairness.

The form of evaluations we review here are quite unique, the ratings from the newspapers would have little effect on the outcomes of footballers, such that the presence of bias would not have a direct negative impact on a player's wages or future outcomes like it would in a workplace (Powell & Butterfield, 1997). This conclusion is supported by the finding that racial pay discrimination is not apparent in Italian football, suggesting that football clubs are not influenced by media representations of black players. However, other issues may arise that are specific to the context of mass sporting media. For example, racial bias within sports newspapers may lead to a spillover effect onto consumers, causing football supporters to potentially develop unconscious preferences against black footballers. Similarly, the use of the newspaper ratings within fantasy football in Italy may create distortions as there is a monetary prize attached to success. Finally, biased results may present an issue to researchers that adopt the newspaper ratings as a measurement for a player's performance within their analysis.[11]

A final implication of the racial bias regards how policies may be implemented to reduce future bias. The authors of the original study, reiterate the suggestion that the bias our results uncover is likely due to unconscious discrimination on the part of the journalists. They follow this by suggesting that unintentional discrimination may be combatted by making the journalists aware of the bias. This follows the findings by Pope et al. (2018), who found that racial bias in foul calls within American basketball (as uncovered by Price and Wolfers (2010)) disappeared after mass media coverage over the study. This is a solid proposal on how to limit bias arising from journalists' evaluations of players, as Pope et al. (2018) highlight that awareness of subtle biases and a willingness to attribute them to internal prejudices are necessary in controlling them. However, it is important to note that other possible mechanisms may have contributed to the finding by Pope et al. (2018), such as players adjusting their behaviour in response to expected bias from referees, or the basketball governing body making unannounced changes to limit bias. Due to this, we are unable to say whether making the journalists aware of the ratings bias would be effective in reducing implicit discrimination. Further, if the bias is due to preference-based discrimination, it is unlikely that the journalists would adjust their evaluations. To overcome this issue, another avenue to reduce bias in the evaluations would be to put more weight upon objective performance metrics when formulating the player evaluations.

To examine the robustness of our baseline findings, we performed several sensitivity analyses. First, to determine whether the bias within ratings was due to the subjectivity of the evaluations, we re-ran the model using an 'objective' measure of player performance provided by www.whoscored.com. We find that the effect of a player's race is insignificant on the measure of player performance, confirming the finding that racial bias within the newspaper ratings is due to subjectivity.

Next, we review the challenge of classifying an individual's race for analysis and test different classifications. First, we determine the extent it matters to distinguish between the number of people who labelled the player as black, by employing a set of dummy variables

[11] For example, Della Torre et al. (2014, 2018) use player ratings from Gazzetta dello Sport to measure the performance of footballers in Italy.

indicating the amount a player's classified as black. We find that across the dummy variables that there is a negative racial bias in ratings, however, it is only statistically significant when three people label the player as back, potentially highlighting a concern with the sensitivity of the variable indicating race. To further review this, I extend on the sensitivity analyses conducted in the original study by adjusting the variable indicating a players race with different thresholds. The results show the significance of racial biases on ratings decreases as we make the classification more stringent. This highlights the challenge of classifying the race of players, suggesting issues with oversensitivity for the original classification which may lead to potentially biased results. Despite this, the presence of racial bias on the average of ratings is still significant until we use the most stringent, suggesting that racial bias in ratings persist even when we account for the sensitivity of the race variable.

The final sensitivity analysis we replicate examines the extent racial bias within newspaper evaluations is influenced by nationality bias. The racial composition of the Serie A complicates our original finding, as there are only 7 players in our dataset who are classified as black Italian, compared to 81 black non-Italians. This may suggest that our finding of racial bias is indicative of favouritism towards Italians instead of bias against black players. By re-estimating our model using a subsample of non-black players, changing our variable of interest to indicate if a player's Italian, we find that favouritism toward Italian players is weakly significant on the rating of players, leading us to conclude that it is not influencing the finding of racial bias. I expand on this analysis by reviewing if racial bias is still present if we filter our analysis to just Italian players. We find that the ratings penalty for black players is still present within this specification, however, it is not significant for any newspaper. This provides suggestive evidence that nationality bias is influencing the findings of racial bias within player ratings, but this is driven by foreign players being penalised instead of Italians seeing a ratings premium. We are cautious, however, in saying this is a definite explanation. Due to the low number of black Italians in our dataset, the statistical power for this analysis is also low; this means that, even if there was an underlying effect of racial bias, we would not detect it in this analysis. Finally, to ensure that racial bias is still present even when we account for potential influences of nationality, we filter the players for our estimation to all non-Italian players. As expected, we find a significant racial bias, albeit less significant than our baseline findings, allowing us to confirm that racial bias is still present within the newspaper ratings of footballers.

The minimal effect we uncover of ingroup nationality bias is somewhat surprising and provides a fair contrast to previous studies that review subjective evaluation bias within similar sporting contexts, such as Krumer et al. (2022), Sandberg (2018), or Zitzewitz (2006). A large difference between the two settings though is that the studies mentioned review contests where competitors are competing against each other for their nations. This may suggest that there is a greater incentive for judges to favour athletes of their own nationality, as success for the competitor may lead to greater support for the sport within their country, or success for their country may create a sense of national pride. Whereas in the present context, there is less apparent incentive to favour domestic players, as football clubs compete against each other instead of nationalities. Similarly, it may be the case that nationalistic bias is not as present in Italian football as it may be in other contexts. There is some evidence to support this, as Farnell et al. (2024) find that there is not a wage premium for Italian players or a wage penalty for international players within the Serie A (although they do find evidence of a pay premium for South American players). However, this only provides small evidence of the absence of nationality bias as it reviews a significantly different context to the focus of our study.

A peculiar finding throughout our analyses is the contrasting estimates for Gazzetta compared to the other two newspapers. In our baseline results, we find that the size of racial bias from Gazzetta is just under half the size compared to the other newspapers. Similarly, within our robustness analysis, the presence of racial bias from this newspaper disappears when we use a different classification of an individual’s race. And finally, when we conduct our analysis using different subsamples, we find that there is a significant ratings premium to Italian players and no effect of racial bias on the sample of non-Italian players.

The authors of the original study noted the peculiarity of the baseline estimate size for Gazzetta, giving two possible explanations why there is a smaller racial bias. First, they suggest that the goodness-of-fit of the newspapers rating model suggests that the journalists make better use of observable data. This is a strong explanation as we found that the subjectivity of the journalists’ evaluations is the cause of such bias, so it is possible that the journalists for Gazzetta generate their ratings by reviewing statistics about player performance, alongside their own opinion from watching the match. However, we earlier suggest that the ratings from Gazzetta suggest a large centrality bias, meaning that the variance of their subjective ratings is much smaller compared to a more objective rating, possibly suggesting a low goodness of fit, as they do not effectively differentiate between the performances of players. The second suggestion is that the composition of the newspaper's readership may be less prone to racially biased content, as the consumers tend to be younger and relatively more educated. However, this is quite speculative, and the mechanism is not particularly clear, as it is unlikely that fans would notice the small bias to the extent that sufficient pressure would be on the journalists not to discriminate (through market pressure or pressure by consumers). As such, this is the least likely explanation.

I contrast the explanations from the original paper by suggesting that the bias in ratings from Gazzetta is due to a nationality bias favouring Italians apart from a racial bias. The first finding that suggests this comes from the analysis of the classification of race within the sensitivity analysis, as the effect of racial bias is not present when we use different race classification, which may mean the baseline finding was inaccurate due to the potential sensitivity of the original variable and not because racial bias is present. Second, the results from the analysis on different subsamples find that there is a statistically significant ratings premium for Italian players from the newspaper, along with no effect of racial bias when we filter our sample to just non-Italian players, providing evidence of national favouritism over racial bias. The finding of national bias instead of racial bias is unexpected when comparing the results to the other newspapers and it is difficult to identify the underlying mechanisms, although it likely highlights that the journalists for Gazzetta have different preferences regarding these characteristics when compared to the journalists at Corriere and Tuttosport. However, this does provide further insights into how bias arises within subjective evaluations and how it can be influenced by nationality bias.

5.2 Areas for Further Research

There are several avenues for future research to explore that will assist in furthering our understanding of bias within subjective evaluations. First, and the simplest addition, would be to expand on the current research with more up to date data, or by reviewing other countries. The data for the present study covers a period from 2009 up to 2018, giving us a rich dataset for analysis; however, making sure the findings are current and using a larger dataset so the results are more accurate will benefit the area. Similarly, this study reviews bias in newspaper ratings just within Italian media. Shifting the focus of the research to a different country will help further the understanding of the generalisability of our results and how bias may differ in other contexts.

A limitation of our research is that we are unable to properly conclude the mechanisms behind the bias we uncover. As discussed within Section 2, much of the empirical literature touching on discrimination has a greater focus on establishing the presence of discrimination over identifying the mechanisms behind it, with just under half of the literature conducting a formal test on the source of discrimination (Bohren et al., 2019). Understanding the mechanisms behind discrimination is important for designing effective policy to limit the presence of bias, however, as the methods of intervention differ significantly depending on the model of bias. For example, with our finding we are unable to fully distinguish between preference-based or implicit discrimination, meaning that we are unable to suggest whether making the journalists aware of the bias or some other intervention would be effective. The authors of the original study touch on this, suggesting a route for future research may investigate the extent to which newspapers implicitly respond to a 'demand for discrimination.' This suggestion provides an interesting avenue research particularly when we are reviewing areas such as media. One possible way we can review this is by quantifying the political leanings of newspapers that cover sport alongside the regular news to understand the extent to which it effects bias within their evaluations of players.

Finally, further examinations of bias within sporting media and any potential consequences provides an interesting area that warrants future research. Specifically, I suggest three topics that provide a strong follow up to the present study. First, we have stated that our setting provides a unique context to study performance evaluations, as there is little expected effect on the present and future outcomes of the players being assessed. However, a thorough examination of this subject will provide interesting insights into how media coverage of individuals affects their earnings. This can be expanded outside of player ratings by examining how media coverage of athletes in general affects their outcomes. Second, future research may examine the effect of biased evaluations outside of the direct effect on athletes. As discussed earlier, the bias within the evaluations may lead to adverse spillover effects, such as fans developing negative preferences against footballers due to their race, or distortions being created within areas such as fantasy football. Due to this, formally testing and understanding if the bias we uncover has adverse outside effects is important. Finally, the ratings we examine that are published by the newspapers are given alongside a small review of each players performance. As such an interesting line of future study may perform sentiment analysis on these small evaluations to understand the extent that they are biased against groups of players.

6. Concluding Remarks

The present study analyses the determinants of footballer ratings in Italian sports newspapers, reviewing if the evaluations of players are influenced by a degree of racial bias; specifically, whether black players receive lower ratings than non-black players, conditional on their performances. We do this by replicating the study by Principe and Van Ours (2022), with some expansions to the analysis to more thoroughly understand the sources of bias and potential methodological issues. Using data from the top division of Italian domestic football, we find that there is such bias within the evaluations of black footballers. This finding broadly contributes towards the literature on discrimination and biases within subjective evaluation, finding similar results to Elvira and Town (2001) in their analysis of employee evaluations in a large U.S. firm, and Price and Wolfers (2010) who review the frequency of foul calls committed by referees within American basketball. We also review if football clubs are biased against black players in terms of wages, however, we do not uncover any evidence of disparities in player wages.

Extending our results, we find that the presence of racial bias within player ratings is focused on the lower end of the ratings distribution, and this is present across newspapers although with varying degrees of bias. Although we are unable to disentangle the mechanisms behind the presence of bias, we suggest that this result shows the disparity in ratings is a result of implicit discrimination on the part of the journalists due to stereotyping. This presents important implications for the design of policies implemented to reduce discrimination, as it suggests that the presence of bias may be reduced by making the evaluators aware of their bias, like how Pope et al. (2018) suggest that bias within foul calls disappeared after the media coverage of a study uncovering bias within basketball foul calls. However, we note that we cannot fully distinguish the bias between implicit discrimination or preference-based discrimination, which may lead to different implications for policy interventions. Finally, expanding the baseline analysis, we find that the presence of bias within one of the three newspapers in review, Gazzetta dello Sport, is driven by nationality bias that favours Italian players instead of racial bias which is suggested by the initial results. We suggest that the baseline results misattribute this ratings differential as racial bias due to the significant majority of black players in the Serie A also being a non-domestic player.

This presence of racial bias presents several concerns for workplace management and the effect mass media has on consumers. First, we suggest that the role of media within society presents unique issues from biased subjective evaluations, particularly the potential spillover effect of bias leading sport newspaper readers to develop adverse preferences against black players. However, we discuss that this effect is likely to be minimal. Within the context of the firm, we appraise the findings of the literature on subjective evaluations by further highlighting that performance evaluations do not necessarily reflect an individual's true quality and are prone to bias, presenting issues with the fairness and efficiency of the allocation of incentives within the workplace and suggesting that for managers reviewing the performance of their employees, they should put more weight on objective measures of their performance.

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Appendix

Table A - Summary Statistics

	Black		Non-Black	
	Mean	Std. Dev	Mean	Std. Dev
Average Rating	5.687	0.578	5.780	0.437
Corriere Rating	5.630	0.763	5.741	0.569
Gazzetta Rating	5.828	0.456	5.884	0.327
TuttoSport Rating	5.603	0.745	5.714	0.598
Log Wage	6.634	0.792	6.644	0.804
Age	27.173	3.942	28.684	4.135
Age-Squared	753.865	218.094	839.870	238.277
Goals	1.398	2.398	2.108	3.773
Assists	1.314	2.116	1.417	2.108
Yellow Cards	3.130	3.155	3.450	3.297
Red Cards	0.184	0.424	0.196	0.448
Shots per Game	0.795	0.805	0.797	0.837
Pass Success %	67.384	31.534	66.263	30.560
Aerials Won	0.795	0.773	0.882	0.854
Tackles	1.467	1.088	1.310	1.027
Interceptions	1.113	0.969	1.109	0.959
Fouls	1.015	0.693	0.965	0.655
Offsides Won	0.161	0.305	0.178	0.333
Clearances	1.789	2.314	1.986	2.373
Dribbled Past	0.523	0.458	0.535	0.461
Blocks	0.186	0.240	0.208	0.250
Key Passes	0.634	0.615	0.610	0.589
Offensive Dribbles	0.750	0.788	0.442	0.489
Fouled	0.915	0.790	0.895	0.712
Offsides	0.101	0.194	0.158	0.313
Bad Controls	0.644	0.595	0.568	0.579
Average Passes	25.945	17.090	26.173	17.345
Crosses	0.311	0.432	0.314	0.430
Long Balls	1.763	1.921	1.814	1.946
Through Balls	0.072	0.123	0.080	0.150
Shots off Target	0.364	0.359	0.345	0.347
Shots on Post	0.011	0.034	0.013	0.037
Shots on Target	0.239	0.313	0.265	0.335
Blocked	0.195	0.219	0.190	0.226
Dispossessed	0.801	0.731	0.682	0.682
Inaccurate Cross	1.122	1.384	1.100	1.315
Inaccurate Control	0.135	0.334	0.163	0.422

Table B - Effect of 'Black' on Newspaper Ratings and Wages - Full Parameter Estimates

	Rating AVG	Corriere	Gazzetta	TuttoSport	Ln Wage
Effect of Black	-0.085 ***	-0.096 **	-0.059 **	-0.102 ***	-0.005
	(0.030)	(0.040)	(0.030)	(0.038)	(0.05)
Age	0.000	-0.024	-0.02	0.042	0.484 ***
	(0.027)	(0.035)	(0.023)	(0.043)	(0.054)
Age-Squared	0.000	0.001	0.000	-0.001	-0.008 ***
	(0.000)	(0.001)	(0.000)	(0.001)	(0.001)
Goals	0.029 ***	0.037 ***	0.020 ***	0.031 ***	0.017 **
	(0.004)	(0.006)	(0.004)	(0.007)	(0.007)
Assists	0.026 ***	0.034 ***	0.021 ***	0.023 ***	0.017 *
	(0.006)	(0.008)	(0.004)	(0.008)	(0.009)
Yellow Cards	0.001	0.002	0.000	0.000	0.002
	(0.003)	(0.004)	(0.003)	(0.004)	(0.005)
Red Cards	-0.043 ***	-0.046 **	-0.046 ***	-0.036 *	-0.003
	(0.015)	(0.020)	(0.012)	(0.020)	(0.022)
Shots per Game	-0.014	-0.018	0.161	-0.186	0.069
	(0.150)	(0.204)	(0.106)	(0.224)	(0.231)
Pass Success %	-0.009 ***	-0.011 ***	-0.004 ***	-0.012 ***	-0.002
	(0.001)	(0.001)	(0.001)	(0.002)	(0.001)
Aerials Won	-0.054 ***	-0.063 ***	-0.034	-0.064 **	-0.025
	(0.019)	(0.022)	(0.022)	(0.027)	(0.029)
Tackles	0.034 *	0.039 *	0.036 **	0.028	-0.006
	(0.018)	(0.023)	(0.016)	(0.024)	(0.027)
Interceptions	0.046 **	0.056 **	0.028 *	0.054 **	-0.029
	(0.018)	(0.027)	(0.015)	(0.026)	(0.026)
Fouls	-0.200	-0.041	-0.01	-0.009	-0.034
	(0.030)	(0.039)	(0.027)	(0.040)	(0.040)
Offsides Won	-0.061	-0.101	-0.038	-0.043	0.136 **
	(0.042)	(0.069)	(0.034)	(0.060)	(0.058)
Clearances	0.033 ***	0.047 ***	0.012	0.039 ***	0.013
	(0.010)	(0.013)	(0.008)	(0.013)	(0.013)
Dribbled Past	-0.033	-0.026	-0.046	-0.027	-0.083 *
	(0.037)	(0.050)	(0.032)	(0.050)	(0.048)
Blocks	0.277 ***	0.309 ***	0.134 ***	0.389 ***	-0.063
	(0.057)	(0.077)	(0.046)	(0.077)	(0.083)
Key Passes	-0.067	-0.116	-0.003	-0.083	0.055
	(0.052)	(0.073)	(0.035)	(0.072)	(0.054)
Dribbles	0.090 ***	0.079 ***	0.066 ***	0.127 ***	0.025
	(0.022)	(0.029)	(0.019)	(0.032)	(0.039)
Fouled	0.029	0.035	0.019	0.033	-0.001

	(0.023)	(0.032)	(0.020)	(0.029)	(0.029)
Offsides	0.017	0.016	-0.094	0.129	0.039
	(0.070)	(0.091)	(0.078)	(0.085)	(0.073)
Bad Control	0.055	0.122 ***	-0.022	0.064	0.005
	(0.034)	(0.046)	(0.034)	(0.050)	(0.040)
Avg Passes	0.013 ***	0.016 ***	0.005 ***	0.017 ***	0.012 ***
	(0.002)	(0.003)	(0.001)	(0.003)	(0.002)
Crosses	0.176 ***	0.162 *	0.183 ***	0.182 **	-0.032
	(0.065)	(0.093)	(0.049)	(0.089)	(0.066)
Long Balls	-0.013	-0.019	0.001	-0.019	0.022
	(0.009)	(0.013)	(0.007)	(0.012)	(0.016)
Through Balls	-0.033	0.049	-0.124 *	-0.024	0.053
	(0.080)	(0.107)	(0.068)	(0.116)	(0.144)
Shots off Target	-0.086	0.02	-0.249 **	0.011	-0.006
	(0.159)	(0.214)	(0.114)	(0.237)	(0.237)
Shots on Post	0.074	-0.138	0.296 *	0.065	0.118
	(0.272)	(0.366)	(0.170)	(0.384)	(0.363)
Shots on Target	0.266 *	0.268	0.082	0.447 **	0.118
	(0.147)	(0.206)	(0.108)	(0.216)	(0.228)
Blocked	0.005	-0.099	-0.096	0.21	-0.034
	(0.165)	(0.237)	(0.117)	(0.238)	(0.245)
Dispossessed	0.045	0.034	0.01	0.092 **	-0.054
	(0.033)	(0.054)	(0.028)	(0.044)	(0.036)
Inaccurate Cross	-0.043 **	-0.035	-0.050 ***	-0.043 *	-0.040 **
	(0.018)	(0.025)	(0.017)	(0.022)	(0.019)
Inaccurate Control	-0.052	-0.05	0.060 **	-0.045	-0.062
	(0.036)	(0.050)	(0.017)	(0.051)	(0.057)

Note: Standard errors in parentheses; Standard errors are clustered at the individual player level; * - ** - ***: Significant at 10 - 5 - 1%